Do not write outside the box

0 2 This question is about electrolysis.

0 2 . 1 Complete the sentence.

Choose the answer from the box.

[1 mark]

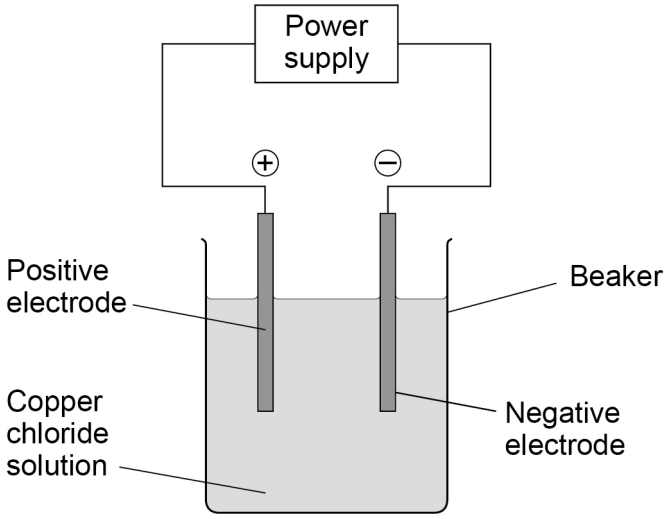
gaseous	**molten**	**solid**

Copper chloride can conduct electricity when in solution or

when _____ .

Figure 1 shows the apparatus used for the electrolysis of copper chloride solution.

Figure 1



There are four ions in copper chloride solution:

- Cu^{2+}
- Cl^{-}
- H^{+}
- OH^{-}

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0 2 . 2 Why do Cl^- ions and OH^- ions move to the positive electrode?

[1 mark]

_____

_____

0 2 . 3 Where do the H^+ and OH^- ions come from in the electrolysis of copper chloride solution?

[1 mark]

Tick (✓) **one** box.

Air ☐

Copper chloride ☐

Water ☐

0 2 . 4 Which ion produces a metal?

[1 mark]

Tick (✓) **one** box.

Cu^{2+} ☐

Cl^- ☐

H^+ ☐

OH^- ☐

Question 2 continues on the next page

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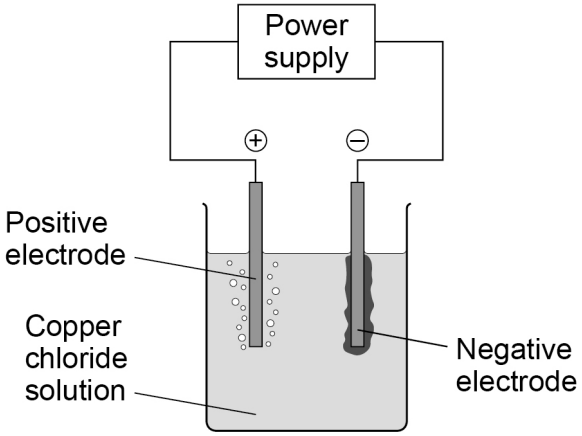
0 2 . 5 **Figure 2** shows the apparatus during the electrolysis of copper chloride solution.

Figure 2



Describe what is seen at each electrode during the electrolysis of copper chloride solution.

[2 marks]

Positive electrode _____

_____

Negative electrode _____

_____

0 2 . 6 500 cm^3 of copper chloride solution contains 6.50 g of copper chloride.

Calculate the mass of copper chloride in 40.0 cm^3 of this copper chloride solution.

[2 marks]

_____

_____

_____

Mass = _____ g

8